



PROGRAMMING WITH PYTHON

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Computer Applications

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Data Ingestion and Multiprocessing

Async IO

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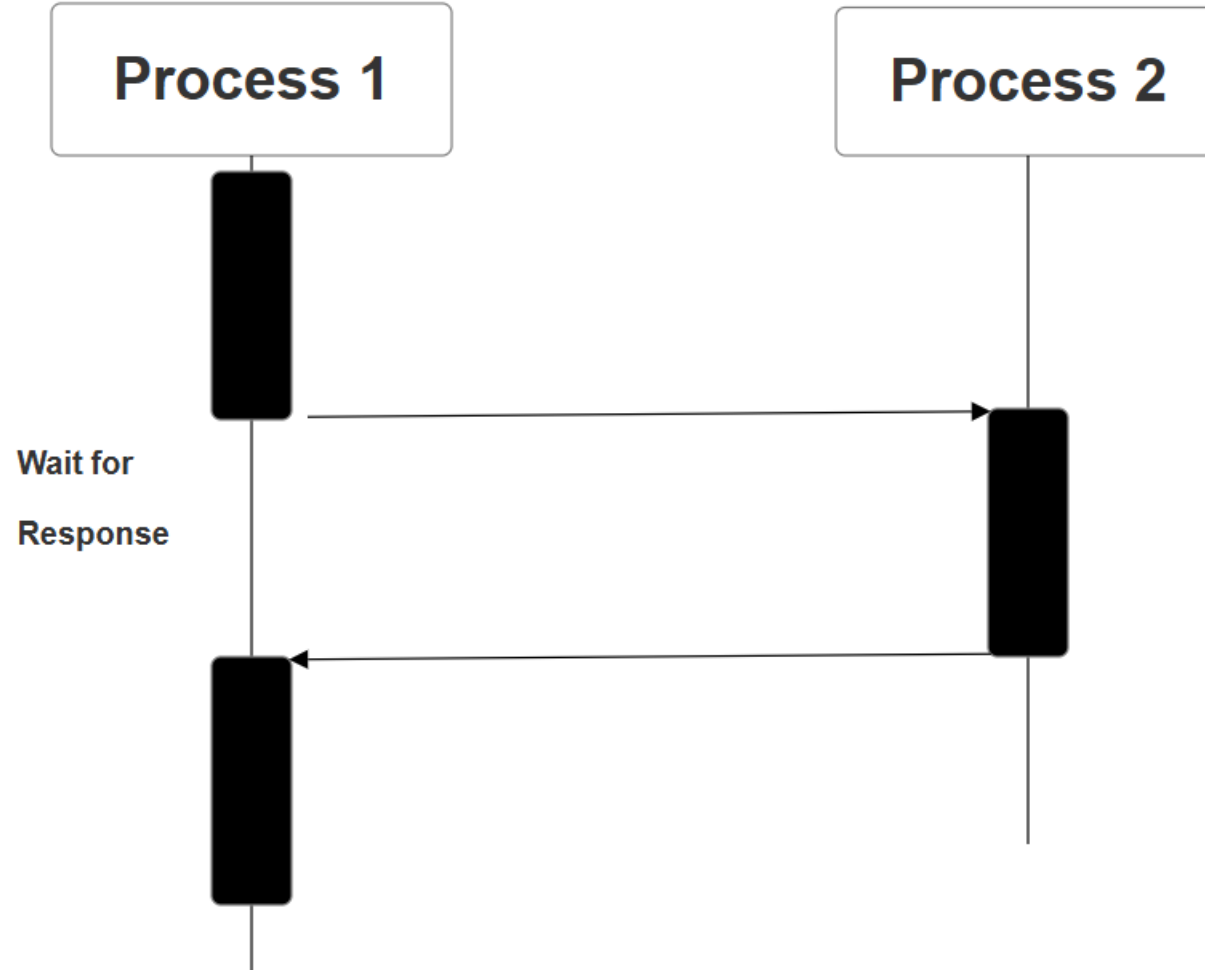
Synchronous Program

- A synchronous program operates by executing one step at a time.
- Programs that feature conditional jumps, loops and function calls can also be regarded as running one step at a time.
- After the completion of each step, the program moves on to the next step.



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Synchronous Program





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Synchronous Programming: Benefits and Drawbacks

- Simplicity
 - Synchronous programming is clear-cut and simpler to implement.
 - Each task is executed in succession, which makes the flow of execution easy to follow and comprehend.



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Synchronous Programming: Benefits and Drawbacks

- Ease of Debugging
 - Debugging synchronous code is more straightforward due to the linear execution flow, resulting in fewer concurrency-related issues, such as race conditions or deadlocks.



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Synchronous Programming: Benefits and Drawbacks

- Predictability
 - The sequential characteristic of synchronous operations guarantees that tasks are performed in a predictable sequence, which is crucial for operations that require strict ordering.



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Synchronous Programming: Benefits and Drawbacks

- Inefficiency with I/O Operations
 - Synchronous programming may prove inefficient for I/O-bound tasks, as actions like file reading or network requests can halt the entire thread until they are completed.



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Synchronous Programming

- Use Cases
 - It is appropriate for simple applications, scripts, or tasks where operations are quick and do not involve extensive I/O or prolonged waiting periods, such as command-line utilities or basic data processing scripts.



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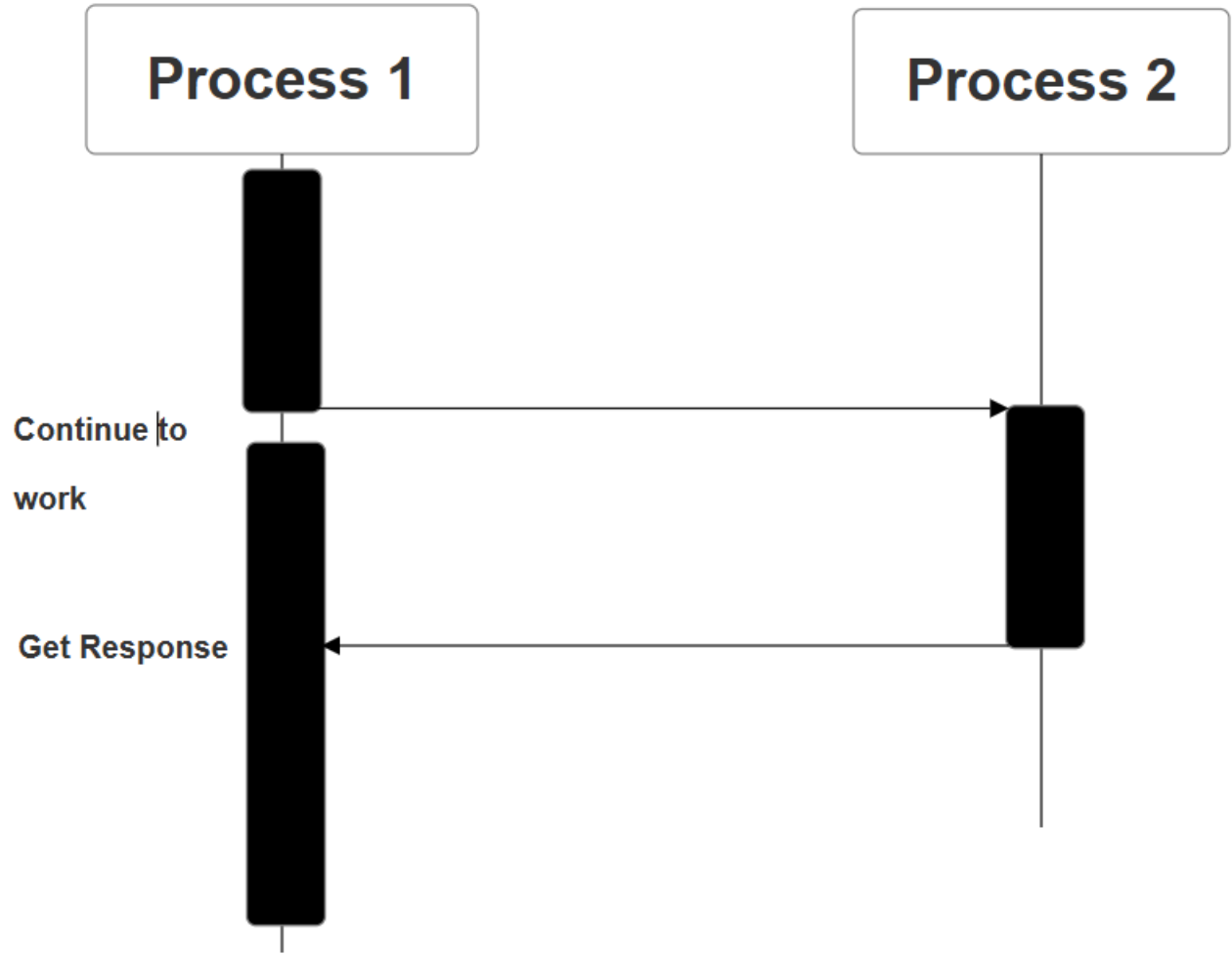
Asynchronous Program

- Asynchronous programming facilitates the independent execution of tasks, which allows for concurrent processing and enhanced performance.
- In contrast to synchronous programming, where each task must wait for the prior task to finish, asynchronous tasks can operate concurrently, making better use of resources and improving the responsiveness of applications.



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Asynchronous Program





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Asynchronous Programming: Benefits and Drawbacks

- Concurrency
 - Asynchronous programming facilitates the execution of multiple tasks simultaneously, enhancing performance and optimizing resource use, particularly in scenarios involving I/O-bound and high-latency operations.



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Asynchronous Programming: Benefits and Drawbacks

- Responsiveness
 - Asynchronous operations allow applications to maintain their responsiveness, which is especially crucial in user interfaces and real-time systems where any delays can negatively impact user experience.



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Asynchronous Programming: Benefits and Drawbacks

- Scalability
 - Asynchronous programming is capable of managing a greater number of concurrent operations, rendering it suitable for scalable web servers and networked applications.



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Asynchronous Programming: Benefits and Drawbacks

- Complexity
 - The implementation of asynchronous code is inherently more complex, necessitating meticulous management of concurrency, which may result in elusive bugs and more difficult debugging processes.



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Asynchronous Programming

- Use Cases
 - It is most effective for applications that require the handling of multiple tasks simultaneously, such as web servers, chat applications or any system that demands real-time interaction and high throughput, including streaming services or asynchronous data processing tasks.



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Recap

- Synchronous Programming
- Asynchronous Programming
- Benefits and drawbacks
- Use cases



THANK YOU

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